



is available on all PC platforms.

We particularly like Stellarium's rewind and forward feature, which helps you visualise the steady movement of the planets, the Milky Way galaxy, and everything related to it. Stellarium supports real-time movement of celestial objects, a catalogue of over 6,00,000 stars.

Using zoom, you can track and locate a galaxy like the M31, also known as the Andromeda Galaxy, which can be seen with a pair of binoculars. Many other spiral galaxies can be seen with a basic telescope.

If you are into living out Star Trek fantasies, albeit for free, then Celestia, a real-time 3-D space simulation software



Celestia, an astronomy software is almost like a game

is a good bet. This software is good enough to give a conventional game a run for its money. You can use it to watch the solar eclipse as it cuts through Earth, travel light years into the deepest parts of the universe, or travel back or forward in time to see the Halley's Comet.

The controls for Celestia are quite intuitive, similar to the space simulator game Homeworld from Relic. It lets switch the scale and viewing angle to zoom in and out of any part of the universe in a flash.

Buying a telescope

The Bangalore Astronomical Society recommends that one buy a telescope not just on the basis of its magnification (often emphasised in packages), and that a good pair of binoculars with good optics would be better than a cheap telescope. Reflector telescopes are cheaper than refractor telescopes. Newtonian

Types of space objects

Star Clusters	Globular clusters M2, M3, M4	Open clusters M45, M11, M18,	Supernova Remnant M1: Crab Nebula
Nebula	M1: Crab Nebula	M8: Lagoon Nebula	M16: Eagle Nebula
	M17: Horseshoe Nebula	M20 Trifid Nebula	M27: Dumbbell Nebula
Galaxies	Spiral galaxies M51, M31, M33, M104	Lenticular galaxies M84, M85, M86	Elliptical galaxies M87, M89

reflecting telescopes, aka table top telescopes are made in India by a company called Startracker systems. The complete kit, which includes eye pieces, lenses, and a tripod, starts from Rs. 4,800 and goes up to Rs. 14,700. While the low-end 75-mm telescope can let you see a 10.5 magnitude star, the high-end 125-mm version can see a 12.5 magnitude star.

An astronomer's get-away guide to India

Finding low-light areas is quite difficult in a country like India. Avoid the big cities, villages, and mapped out in bright dots in this light map picture courtesy NASA.

Forest reserves, hill stations, deserts, coasts, islands, and long treks generally lead you to a place where you can see a lot of stars.

NASA images

In 1966, Stuart Brand, best known as the editor of the Whole Earth Catalog, campaign to have pictures of Earth as seen from space available in the public domain. Since 1968, NASA has been a leading supplier of high-definition astronomy pictures available through a network of web sites. You can download poster size pictures of the universe, solar system, satellite photos and visualisations of Earth.

NASA images is a partnership between NASA and the Internet archive, a non-profit digital library based out of the US. The content is without copyright and can then be reproduced, as long as image credits are given.

You can download the iconic image known as the Blue Marble, a light map of the world that is 27 MB in size, and contain photographs from space probes like the Voyager and photos from the outer edge of civilisation from the International space Station. Spiral galaxies and nebulae courtesy Hubble are also available. The thousands of pictures that show up in a search "moon", for example, can be narrowed down quickly, by year, the space laboratory and subject matter.

Messier objects

Seventeenth century French astronomer, Charles Messier was notable for publishing a catalogue of deep space objects such as nebula star clusters that came to be known as the 103 Messier objects, which includes nebula, galaxies, clusters, and supernova remnants. It was published to help other astronomers distinguish between permanent and transient objects in the sky. For amateur astronomers, the Messier collection encompasses some of the grandest objects in the night sky. Many of which, are a delight to see through a telescope the first time.

According to Wikipedia, the human eye can pick up faint stars up to +6.5 magnitude under a modern dark sky.

These Messier objects all have an apparent magnitude equal to or below that. The following should be easy to spot and watch out for: The Butterfly Cluster (M6) and Ptolemy Cluster (M7), Eagle Nebula (M16), The

Horseshoe Nebula (M17), Trifid Nebula (M20) Sagittarius Cluster (M22), the Andromeda Galaxy (M31), Triangulum Galaxy (M33), Orion Nebula (M42), and the Pleiades cluster (M45). 

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Ascension: the rising of a star above the horizon
Inclination: the angle between the plane of the orbit and the plane of the ecliptic stated in degrees
Azimuth: The azimuth of a celestial body is the angle between the vertical plane containing it and the plane of the meridian